

Quiz (Carolina Reaper)

- Q1.** A 10-ounce solution is 30% alcohol. How much alcohol is in the solution?
(A) 2 ounces
(B) 3 ounces
(C) 4 ounces
(D) 5 ounces
(E) 6 ounces
- Q2.** What is the ratio of water to alcohol in a 15% alcohol solution?
(A) 1 : 1
(B) 1 : 2
(C) 2 : 1
(D) 4 : 1
(E) 6 : 1
- Q3.** If a 20-ounce mixture of juice and water is 25% juice, how many ounces of juice are in the mixture?
(A) 4 ounces
(B) 5 ounces
(C) 6 ounces
(D) 8 ounces
(E) 10 ounces
- Q4.** A 5-ounce solution of acid is 20% nitric acid. How much nitric acid is in the solution?
(A) 0.5 ounces
(B) 1 ounce
(C) 2 ounces
(D) 3 ounces
(E) 4 ounces
- Q5.** A coffee shop sells a 12-ounce coffee that is 80% coffee and 20% cream. How many ounces of coffee are in the 12-ounce mixture?
(A) 8 ounces
(B) 9.6 ounces
(C) 10 ounces
(D) 10.4 ounces
(E) 12 ounces
- Q6.** What is the concentration of a 15-ounce mixture that contains 3 ounces of solute?
(A) 10%
(B) 15%
(C) 20%
(D) 25%
(E) 30%
- Q7.** A pharmacist mixes 20 milliliters of a 40% solution of medication with 30 milliliters of a 20% solution. What is the concentration of the final mixture?
(A) 20%
(B) 25%
(C) 28%
(D) 30%
(E) 32%
- Q8.** If 5 liters of a 15% salt solution is mixed with 10 liters of a 25% salt solution, what is the concentration of the final mixture?
(A) 15%
(B) 18%
(C) 20%
(D) 22%
(E) 25%

Open Ended Questions (FRQ)

- Q9.** A chemist has two solutions of acid: one is 30% acid and the other is 70% acid. How many liters of each should be mixed to obtain 20 liters of a 50% acid solution?
- Q10.** A 40-ounce mixture of water and juice is 60% juice. If 10 ounces of water is added to the mixture, what is the new concentration of juice in the mixture? Show your work and explain your reasoning.

Chef's Tips

- Read each question carefully
- Use the space provided for your work
- Show all work and explain reasoning for free-response questions